lower than that of red are also not visible, and correspond to infrared radiation.) In fact, as the frequency of the light used is decreased, there comes a point when the speed of the emitted electrons drops to zero and they stop being ejected from the surface, regardless of the possibly blinding intensity of the light source. For some unknown reason, the color of the impinging light beam—not its total energy—controls whether or not electrons are ejected, and if they are, the energy they have.

To understand how Einstein explained these puzzling facts, let's go back to the warehouse, which has now heated up to a balmy 80 degrees. Imagine that the landlord, who hates children, requires everyone under the age of fifteen to live in the sunken basement of the warehouse, which the adults can view from a huge wraparound balcony. Moreover, the only way any of the enormous number of basement-bound children can leave the warehouse is if they can pay the guard an 85-cent exit fee. (This landlord is such an ogre.) The adults, who at your urging have arranged the collective wealth by denomination as described above, can give money to the children only by throwing it down to them from the balcony. Let's see what happens.

The person carrying pennies begins by tossing a few down, but this is far too meagre a sum for any of the children to be able to afford the departure fee. And because there is an essentially "infinite" sea of children all ferociously fighting in a turbulent tumult for the falling money, even if the penny-entrusted adult throws enormous numbers down, no individual child will come anywhere near collecting the 85 he or she needs to pay the guard. The same is true for the adults carrying nickels, dimes, and quarters. Although each tosses down a staggeringly large total amount of money, any single child is lucky if he or she gets even one coin (most get nothing at all) and certainly no child collects the 85 cents necessary to leave. But then, when the adult carrying dollars starts throwing them down—even comparatively tiny sums, dollar by single dollar—those lucky children who catch a single bill are able to leave immediately. Notice, though, that even as this adult loosens up and throws down barrels of dollar bills, the number of children who are able to leave increases enormously, but each has exactly 15 cents left after paying the guard. This is true regardless of the total number of dollars tossed.